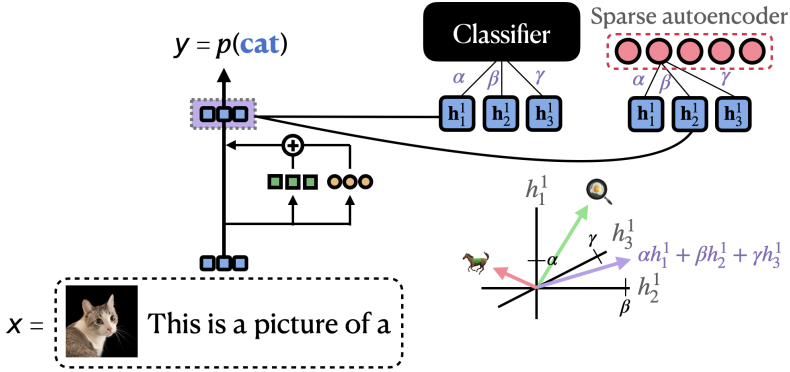


**Figure 4**

Neurons are not guaranteed to encode interpretable features. If non-basis-aligned directions encode the features of interest, then a neuron may activate on many different features that are non-orthogonal to its basis. Locating non-basis-aligned mediators requires components in addition to the model’s computation graph that encode the coefficients on each activation. One can, for example, obtain these coefficients via supervised optimization with probing classifiers (§6.2.1) or unsupervised optimization with sparse autoencoders (§6.2.2). Note that optimization-based techniques sometimes introduce non-linearities, meaning that the discovered directions will not necessarily be a subspace of activation space.

$h_i^\ell$ . If we multiply this slope by the difference in  $h_i^\ell$  and a counterfactual value  $h_i^{\ell'}$ ,<sup>26</sup> then we can obtain a linear approximation of how much changing  $h_i^\ell$  would have changed  $m$ —in other words, a linear approximation of the indirect effect (Eq. 1). We can perform this attribution for all  $h_i^\ell$  in parallel using  $O(1)$  forward and backward passes.

The second and more difficult disadvantage to overcome is that exhaustive search constrains us to finite sets of mediators. Thus, this approach will not be possible if the search space is continuous (infinitely large). This is a key motivation behind the methods in the following subsection.

## 6.2 Optimizing over large spaces of mediators

For some types of mediators, the collection of candidate mediators is continuous or far too large to exhaustively search over; this precludes using methods described in §6.1. To search over large but enumerable sets, some researchers employ modified versions of exhaustive search, including greedy search methods (Vig et al. 2020) or manual searches (Wang et al. 2023). For continuous spaces, however, interpretability researchers generally use optimization. We taxonomize these optimization problems based on whether they require the interpretability researcher to manually select and incorporate task-specific information into the loss function (supervised methods; §6.2.1) or not (unsupervised methods; §6.2.2). We illustrate the intuition behind optimization-based search in Figure 4.

**6.2.1 Supervised mediator searches.** By *supervised mediator searches*, we mean parametric approaches that require labeled task data  $\mathcal{D}$  and/or human-provided hypothesized

<sup>26</sup> A common counterfactual value includes  $\mathbb{E}_{x \in X} [h_i^\ell]$ ; setting  $h_i^\ell$  to this mean is known as a mean ablation. Setting  $h_i^\ell$  to 0 is known as a zero ablation. See Appendix A for more details.

high-level causal graphs  $\mathcal{H}$ . For example, these methods might require the researcher to propose candidate intermediate concepts  $V_i \in V$  that they expect the model to use in performing some task, or a candidate mechanism (a subset of  $\mathcal{H}$ ) by which the model might complete the task. Others might simply require labeled data  $(x, y) \in \mathcal{D}$  for training classifiers.

*Supervised probing.* In supervised probing approaches, the researcher hypothesizes that the model represents some concept  $V_i \in V$ , designs some labeled dataset consisting of  $\{(x_i, y_i)\}_{i=1}^n \in \mathcal{D}$  to isolate the concept, and trains a probe to locate some direction(s) in  $Z$  that *correlate(s)* with the concept. A probe is generally formulated as a classifier  $\Pi$  that maps from model representations  $Z$  given input  $x_i$  to probabilities over class predictions:  $p_\Pi(y_i | x_i, Z)$ . Generally,  $Z$  is a representation vector  $\mathbf{h}^\ell$  at the end of a particular layer  $\ell$ , and the probe updates its weights over all possible subspaces therein to attend to signals that are predictive of the labels. Many papers employ probing classifiers (Belinkov and Glass 2019), though most do not validate the causal efficacy of the probe (Belinkov 2022). A drawback of this is that certain activation subspaces are often *correlated* with a concept without *causally mediating* the concept. This means that probing can return many false positives (Hewitt and Liang 2019; Elazar et al. 2021; Ravichander, Belinkov, and Hovy 2021; Amini et al. 2023; Belinkov 2022).

Thus, much recent work complements supervised probing approaches with additional checks of causality. For example, one can apply causal mediation analysis to the directions  $W_\Pi^\top \mathbf{h}^\ell$  identified by supervised probing (Marks and Tegmark 2023; Nanda, Lee, and Wattenberg 2023)—i.e., one can measure whether the IE (Eq. 1) of the direction used by the probe is significantly different from zero. This would fall under non-basis-aligned mediators. One can also backpropagate from the classifier to modify the behavior of the model (Giulianelli et al. 2018) or to generate counterfactual representations (Tucker, Qian, and Levy 2021), which is more akin to a full-layer intervention. One can also directly compare the probe’s predictions to a causally grounded probe (Amini et al. 2023).

Another line of work uses the directions discovered by probes to *guard* or *erase* information about a particular concept from the model’s representations. For example, a direction in a model’s activation space that is most predictive of the target concept can be nullified via orthogonal projections, such that the model can no longer use the information (Ravfogel et al. 2020); this process can be repeated until linear guarding is achieved. The aim of this procedure is to make the indirect effect of the concept on the model’s behavior equal to zero for all inputs.

Concept erasure and guarding can be used to measure the causal importance of particular concepts, as in Elazar et al. (2021), though studies that employ methods like these tend to focus on single layers. More recently, techniques such as LEACE (Belrose et al. 2023) and follow-ups (Singh et al. 2024b) have generalized this idea to provably prevent any linear classifier from using a concept; this moves beyond orthogonal projections and projects out the information at *every* layer. One could use such methods to causally understand the *set of* directions, or non-basis-aligned multidimensional space, that encode some concept. Note that many of these methods are still susceptible to the problems entailed by using linear mediators; thus, future work could follow Iskander, Radinsky, and Belinkov (2023); Ravfogel et al. (2022) in generalizing these analyses to non-linear mediator types.

*Counterfactual-based optimization.* A related line of methods involves learning binary masks over sets of components (e.g. sets of neurons, attention heads, layers) to deter-

mine which are relevant mediators for a task. Examples include subnetwork probing (Cao, Sanh, and Rush 2021) and Desiderata-based Component Masking (DCM) (Davies et al. 2023; Prakash et al. 2024). These allow us to find basis-aligned multidimensional subspaces (§5.2), but they require triplets  $(x_i, x'_i, y_i)$  consisting of contrastive input pairs  $(x_i, x'_i)$  and a target  $y_i$ .

A more expressive class of approaches involves using the result of causal mediation analysis as a metric to directly optimize. One such line of methods includes Distributed Alignment Search (DAS) and follow-up methods such as Boundless DAS (Geiger et al. 2024; Wu et al. 2023; Huang et al. 2024). These methods are powerful in that they allow us to locate non-basis-aligned multidimensional subspaces (§5.3) that correspond to a particular concept, though they have relatively strict data requirements compared to probing: we need not only labeled examples, but also contrastive input pairs *and* a human-provided high-level causal graph. More precisely, given an activation vector  $\mathbf{h}^\ell$  and hypothesized causal graph  $\mathcal{H}$  containing nodes and edges  $(V, E)$ , the goal is to learn an invertible linear transformation  $\mathbf{R}$  such that a concept of interest  $V_i \in V$  is aligned to the bases of the transformed space  $\mathbf{R}(\mathbf{h}^\ell)$ . To locate the concept, one iterates over contrastive input pairs  $(x_i, x'_i)$  in training dataset  $\mathcal{D}$  that vary only with respect to  $V_i$  and (ideally) no other concepts. For each pair, interventions are performed in the transformed space such that  $v_i$  (the value of  $V_i$  given  $x$ ) is set to a counterfactual value  $v'_i$  that it would have taken given  $x'_i$ . Finally, the space is transformed back to the original space via  $\mathbf{R}^{-1}$ . This should result in a predictable change in model behavior, as defined by  $\mathcal{H}$ . See Appendix C for an illustration and more detailed description.

These methods provide time-efficient ways to search for human-interpretable variables encoded in intractably large or innumerable mediator sets. However, their key limitation is that they require either labeled data, contrastive pairs of inputs, and/or a pre-existing causal hypothesis as to how a model accomplishes some behavior. These methods can be evaluated with respect to accuracy in capturing model behavior, but they do not directly indicate *a priori* what those hypotheses should be. When one obtains negative results, these methods also do not indicate in what specific ways the hypotheses are wrong. They also require sufficient training data to recover the concept of interest during training. As with all parametric methods, the above approaches are subject to overfitting or underfitting.

**6.2.2 Unsupervised mediator searches.** Supervised search methods (see §6.2.1) require specific hypotheses about the internal representations of neural networks. However, neural networks implement various behaviors, many of which may be counterintuitive to humans and therefore more likely to be missed in supervised settings. For example, while Li et al. (2023) hypothesized a constant board state representation in a Transformer learning to play Othello, Nanda, Lee, and Wattenberg (2023) later found that the model actually switches the board state representation with every turn, taking the view of “my pieces vs. opponent’s pieces” rather than “black pieces vs. white pieces”. This example demonstrates that it can be desirable to search for mediators without specifying beliefs as to what those mediators do ahead of time.

Hence, some studies employ *unsupervised* methods. Typically, unsupervised methods return large—but finite—collections of mediators. Unsupervised methods are largely *correlative*, meaning that the discovered mediators may not necessarily capture causally relevant or faithful aspects of a model’s computation. However, the discovered mediators can then be implicated in a model’s computation post hoc by employing additional techniques, such as exhaustive searches for the highest-indirect-effect mediators (§6.1), to select task-relevant mediators from this collection.